

Mohammed Zakaria Mutlaq

Data Engineer

mohamedzakaria.cs@gmail.com | +20 150 101 5701 | Giza, Egypt | Military: Exempted
[linkedin.com/in/eng-zakaria](https://www.linkedin.com/in/eng-zakaria) | github.com/Eng-Zakaria | eng-zakaria.medium.com | eng-zakaria.github.io/Portfolio

SUMMARY

Data Engineer with 2 years building and operating batch and real-time pipelines with Python, SQL, Spark, and Airflow. Experience covers dimensional modeling, ETL and ELT, data warehouses, and retrieval systems on AWS with Azure exposure. At Waffarha, scaled a bilingual support chatbot into a full customer service agent that investigates cases against live data, escalates with structured handoffs, and serves text and voice calls. Also builds data automation and web scraping pipelines that collect and reconcile data from many sources into reporting-ready tables.

PROFESSIONAL EXPERIENCE

Data Engineer Intern, Waffarha

07/2026 to Present
Cairo

- Scaled a bilingual Arabic and English support chatbot into a full customer service agent that owns requests end to end, checks orders, offers, wallet, and account records in MySQL and ClickHouse, and escalates to the human team with a structured case handoff when confidence is low or the case is sensitive.
- Extended the same agent to voice calls using speech to text input and text to speech output on the existing retrieval and action pipeline, with Redis-backed sessions for concurrent users. Holds a 96.4% pass rate across 110+ evaluation cases with CI regression gating.
- Shipped production retrieval after comparing 4 multilingual embedding models and 5 vector stores on 18,000+ documents. Selected multilingual-E5-large with FAISS at 94.9% recall and 0.62 ms average query latency. Hybrid intent routing answers about 75% of queries with sub-100 ms direct-answer shortcuts before LLM generation.
- Added claim-level fact checking that extracts prices, discounts, and dates from generated answers and removes unsupported claims. Tuned a compact 2 GB MoE model to 88+ tokens per second for reliable on-prem inference.
- Automated ingestion with Airflow DAGs from MySQL and ClickHouse using content-hash dedup, SCD2 versioning, and re-embedding triggers. Containerized FastAPI, Streamlit, Airflow, vector DB, and Redis with Docker Compose for one-command deployment.
- Built data automation and web scraping pipelines: partner price-integrity checks on a daily ingest, scrape, match, compare, and report cycle with BeautifulSoup and Playwright plus per-partner checkpoints and HTML, Excel, and Markdown reports; market-intelligence collectors across app stores, Reddit, News, and web search; merchant-lead discovery from social platforms and Google Maps.

Big Data Engineering Intern, National Telecommunication Institute (NTI)

04/2026 to 07/2026
Smart Village

- Built Spark, Python, and SQL ETL orchestrated with Airflow on Hadoop (HDFS, MapReduce, YARN) with Hive, HBase, and InfluxDB. Built Kafka, Flume, and Sqoop ingestion and AWS (EC2, S3, IAM, VPC) pipelines with Power BI, Tableau, and Grafana dashboards.

Data Engineer, Nexlify Startup

01/2025 to 01/2026
Cairo

- Built an Airflow ETL from 10+ clinics (Excel, clinic APIs) into a centralized warehouse designed from scratch with star schemas and fact, dimension, and bridge tables. Reporting cycles improved 40%.
- Refactored SQL with CTEs and tuned joins across PostgreSQL, SQL Server, MySQL, and SQLite, cutting query latency 30% and clearing BI dashboard timeouts. Cleaned millions of mixed Arabic and English records with Python before load.
- Shipped a single Power BI model with a clinic switcher serving every clinic and company-wide views, replacing per-clinic reports.

AI Instructor, ISchool

06/2024 to 11/2024
Cairo

- Taught data and ML labs to 100+ students with a 90% project completion rate and full curriculum delivery on schedule.

EDUCATION

Bachelor's in Computers and Artificial Intelligence, Helwan University

09/2020 to 05/2024
Cairo

GPA: 3.3 | Graduation Project Grade: A+

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, Scala, Bash

Processing: Apache Spark, Databricks, Flink, Hadoop (HDFS, YARN, MapReduce)

Ingestion and Streaming: Kafka, NiFi, Sqoop, Flume, RabbitMQ, Apache Beam

Orchestration: Airflow, dbt, Dagster, Prefect, AWS Glue

Scraping and Automation: BeautifulSoup, Playwright, rapidfuzz, REST APIs, SCD2, Great Expectations

Cloud and DevOps: AWS (S3, IAM, EC2, VPC), Azure (Blob, Data Factory, Event Hubs, Key Vault), Microsoft Fabric, Docker, Kubernetes, Terraform, GitHub Actions, Linux

Warehousing: Snowflake, Redshift, BigQuery, Delta Lake, Iceberg, Hive, ClickHouse, Star and Snowflake Modeling

Databases and Search: PostgreSQL, MySQL, SQL Server, InfluxDB, Neo4j, Elasticsearch, FAISS, Qdrant

BI and AI: Power BI, Tableau, Grafana, MLflow, PyTorch, XGBoost, FastAPI

TRAINING AND CERTIFICATES

- Huawei HCIP Big Data (H13-723), National Telecommunication Institute (NTI), 2026
- AWS Certified Cloud Practitioner, 2026
- ITI Full Stack Development Trainee, 2021: shipped 3 end-to-end C# and .NET web demos from schema design to UI.

PROJECTS

Telecom Enterprise Data and Automation Platform

Fabric, Azure, dbt, Spark, Terraform, Great Expectations

- Built one medallion Delta Lake platform serving 8 telecom domains with dbt and Spark transforms, local-cloud parity for offline testing, Key Vault only secrets, and dev to staging to prod promotion with rollback. Governed with Purview lineage and Unity Catalog.

Credit Risk Analysis

XGBoost, LightGBM, CatBoost, PyTorch, Airflow, MLflow, FastAPI

- Stacked 4 model families into an ensemble at 0.88 AUC-ROC, orchestrated with Airflow and served with FastAPI, MLflow tracking, Redis lookups, and drift monitoring.

Aircraft Tracking Pipeline

Spark, Kafka, Flume, InfluxDB, Grafana, HDFS, Docker

- Streaming pipeline from live ADS-B data with Spark Structured Streaming into InfluxDB and HDFS, with live Grafana geomaps and per-type aggregates.